



A Novel Framework for temporal knowledge graph reasoning with complex causal relations

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ABSTRACT

In temporal knowledge graph reasoning, existing models face significant challenges in capturing complex causal relationships. When dealing with long-term causal dependencies and cold-start problems, traditional models struggle to effectively capture causality, resulting in decreased inference accuracy and stability. To overcome these shortcomings, we introduce the Latent History Learning for Temporal Causal Graph Attention Network (HTCGAT), which comprises two core modules: Latent History Learning and Temporal Causal Graph Attention Network (TCGAT). Latent History Learning integrates entity similarity and temporal decay to learn latent history information of cold-start entities, effectively mitigating the challenge of capturing historical semantics for newly emerged entities. TCGAT captures entity-level causality and temporal dynamics by using Granger causality to identify causal relations between entities and employing the Temporal Graph Attention Network (TGAT) to model temporal dependencies among events. Comprehensive experiments, including comparative assessments, ablation studies, and case analyses, validate that HTCGAT significantly enhances both inference performance and stability, underscoring its practical applicability in real-world temporal reasoning tasks.

1. Introduction

Knowledge Graphs (KGs), as structured semantic networks, have been extensively employed across diverse domains, such as intelligent question-answering systems (Cui et al. (2023)), recommendation engines (Fan et al. (2022)), and policy development (Lu et al. (2024)). However, static graphs are often unable to capture temporal dynamics in real-world scenarios, where entity states and their relationships continuously evolve. To address this limitation, Temporal Knowledge Graphs (TKGs) have been increasingly recognized as a crucial research domain. In TKGs, knowledge is formally represented through quadruples (s, r, o, t) , which incorporate not only the subject s , relation r , and object o , but also a timestamp t that encodes the temporal evolution of relationships. This dynamic structural representation allows novel approaches to be developed for downstream applications. A persistent challenge, however, lies in the incompleteness of TKG quadruples, with missing of subjects s , objects o , or relations r . Consequently, TKGs reasoning has emerged as a pivotal research problem (Liang et al. (2022)).

TKGs Reasoning (Dasgupta et al. (2018)) is characterized by the deduction of new facts from existing data and can be categorized into two primary paradigms: interpolation and extrapolation. Interpolation is employed to address missing history facts, whereas extrapolation is utilized for future fact prediction. Although history knowledge graphs are enhanced through interpolation by improving their completeness and accuracy for downstream applications, extrapolation is generally considered more significant due to its predictive capabilities (Wang et al. (2023)). This research concentrates on TKG Reasoning in the extrapolation setting, with the objective of predicting absent entities in future timestamps t . Formally, given an incomplete quadruple $(s, r, ?, t)$, the task entails inferring the missing object o .

In reality, events typically exhibit **causal relationships**, where preceding events directly shape subsequent outcomes. This phenomenon is illustrated in Fig. 1 by the quadruple prediction task (*North Korea*, Consult, $?, t$) on the ICEWS18 dataset. A sequence of agreements and disputes from $t - 3$ to $t - 1$ forms a causal chain that leads to a consultation at time t . Causal inference is hindered by **insufficient**

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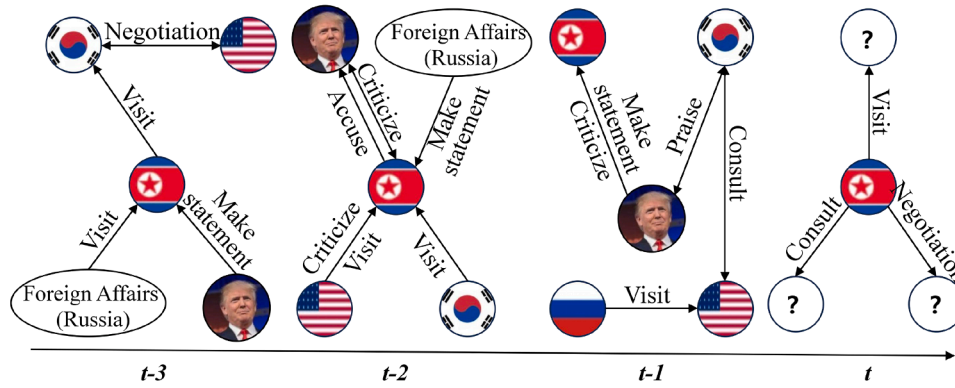


Fig. 1. Example Temporal Knowledge Graphs. Each edge or interaction between entities is associated with temporal information and a set of interactions at a time to construct a multi-relational graph, and the task of extrapolation is to predict future interactions.

historical data for cold-start entities in TKGs, posing a significant challenge. The query (*North Korea*, Visit, $?$, t) identifies Russia as the answer, though lacking direct historical ties. By leveraging entity similarity with the Russian Foreign Ministry, latent history is inferred, enabling discovery of the potential causal link behind North Korea's reciprocal visit at time t .

Recent TKGs reasoning advances focus on structural and temporal representations for future event prediction. Structural features and temporal dependencies are encoded by methods such as RE-GCN (Li et al. (2021)) and RENET (Jin et al. (2020)) through graph neural networks and recurrent architectures. Assumptions of fixed temporal intervals limit their ability to model real-world event sequences with irregular time gaps. Diverse temporal dynamics are better handled by recent models such as GHNN (Han et al. (2020b)) and TECHS (Lin et al. (2023)), where graph attention mechanisms are employed. Entity correlations and statistical patterns are captured, but critical causal relationships are often missed. The importance of causal analysis in TKG reasoning has been increasingly recognized, as shown by approaches like HTCCN (Chen et al. (2024)). In this framework, causal links between entities are uncovered by combining Hawkes processes with temporal convolutional networks to improve predictive reasoning. Fundamental limitations are posed by the model's restricted receptive field and error propagation, resulting in reduced effectiveness for global reasoning and multi-step inference.

To overcome these limitations, we propose the Latent History Learning for Temporal Causal Graph Attention Network (HTCGAT). The architecture uses an event encoder based on Relational Graph Convolutional Networks (RGCNs) to generate timestamped entity representations. Latent History Learning module combines entity similarity and a time decay factor to supplement history context for cold-start entities and enable causal dependency extraction. Temporal Causal Graph Attention Network (TCGAT) embeds causal weights and temporal functions into a Graph Attention Network (GAT) to enhance the framework's ability to capture temporal dynamics and causal structures simultaneously, by analyzing causal relationships between entities using Granger causality analysis (Yin and Barucca (2022)). A decoder computes the joint probability of query events and candidate entities to generate final predictions. In summary, the contributions of this paper are as follows:

- We design a Latent History Learning module to capture potential history interactions for cold-start entities, thereby mitigating the difficulty of capturing causal relationships.
- We propose a temporal causal attention mechanism that integrates causal weights and temporal functions into the graph attention network to represent history fact sequences with varying temporal intervals, causal dependencies among facts, and the dynamic evolution of facts.

- Following extensive experimental validation, the proposed method is shown to effectively mitigate the cold-start problem and demonstrate superior inference performance in the causal modeling of complex time-series data, significantly outperforming existing benchmark methods. These results underscore the method's substantial potential and advantages for practical applications.

2. Related work

In recent years, research on temporal knowledge graph reasoning has made significant progress, advancing the development of artificial intelligence in dynamic knowledge modeling and reasoning. Existing extrapolation models can be broadly categorized into three types: representation-based methods, RNN-based methods, and attention-based methods.

Representation-based approaches (Leblay and Chekol (2018); Ma et al. (2019); Lacroix et al. (2020); Sadeghian et al. (2021); Han et al. (2020a)) are primarily characterized by the employment of embedding techniques, tensor decomposition, and probabilistic modeling to infer temporal evolution patterns of entities and relationships within knowledge graphs. In the study by Leblay and Chekol (2018), relationship validity was derived and temporal dynamics were captured through an effective temporal model. Similarly, a knowledge graph embedding model was introduced by Ma et al. (2019), where low-dimensional vectors are utilized to represent dynamically changing relationships. For TKGs completion, a tensor decomposition-based method was developed by Lacroix et al. (2020), in which high-dimensional data are reformulated into tensor format for efficient processing. The ChronoR model was proposed by Sadeghian et al. (2021), where temporal relationship representation is enhanced through rotation operations. Additionally, complex temporal evolution and nonlinear dependencies are modeled in the DyERNIE model (Han et al. (2020a)) through Riemannian manifold embedding. Despite these advances, higher-order dependencies are not adequately captured by such models, and their flexibility and generalizability in real-world applications are restricted by their reliance on manually designed interpretability mechanisms.

With the advancement of graph neural network (GNN) technology, temporal sequence processing has been further categorized into recurrent neural network (RNN)-based and attention-based methods. In RNN-based approaches (Li et al. (2022a); Tang et al. (2024); Sun et al. (2021); Xu et al. (2023); Zhang et al. (2023)), architectures such as RNNs, LSTMs, and GRUs are predominantly employed for time-series modeling. For instance, in the TIRGN model (Li et al. (2022a)), local and global history patterns are integrated to enhance temporal mapping. The TimeTraveler framework (Sun et al. (2021)) utilizes reinforcement learning to dynamically adjust inference processes, through which accuracy is significantly improved. Similarly, in the CENET model (Xu et al. (2023)), temporal inference is

strengthened through history comparisons, while in the HGSL model (Zhang et al. (2023)), complex temporal variations are represented through joint learning of long- and short-term dependencies. Additionally, event relationships and temporal dependencies are modeled in the DHyper model (Tang et al. (2024)) using dual hypergraphs and RNNs. However, several challenges are faced by these methods, including difficulties in multi-hop relationship modeling, imbalance between local and global information, and adaptation to dynamic graph structures, while sufficient interpretability is often lacking.

Attention-based methods (Han, Chen, Ma, (2021); Han et al. (2020b); Lin et al. (2023); Dong et al. (2024)) are characterized by the integration of graph structural and temporal information for inference. In these approaches, a subgraph-based method (Han, Chen, Ma, (2021)) was developed to enhance transparency, while event-based temporal dependencies are modeled in the Graph Hawkes Neural Network (GHNN) (Han et al. (2020b)) through the combination of GNNs with Hawkes processes. Extrapolative reasoning and interpretability are improved in the TECHS model (Lin et al. (2023)) through the employment of temporal logistic graph networks. Similarly, interpretability is enhanced in the TIP-NN framework (Dong et al. (2024)) by utilizing temporal path learning and inductive reasoning. Although complex graph structures and temporal dependencies are effectively captured by these models, their inference processes are still limited by insufficient transparency and interpretability, which is caused by the lack of explicit causal mechanisms. To overcome these limitations, a shift toward causal relationship modeling has been made in recent research. In the HTCCN model (Chen et al. (2024)), causal inference is advanced through the integration of Hawkes processes with temporal convolutional networks, demonstrating strong short-term prediction performance. However, the model's application to global modeling and complex multi-step prediction tasks is still restricted. With the development of large language models, we also consider TV-LLM proposed by Pan et al. (2025). This approach uses LLMs to generate temporal rules and improves the accuracy of temporal knowledge graph reasoning.

3. Proposed method

3.1. Task definition

Temporal Knowledge Graphs (TKGs) are represented as a sequence of subgraphs ordered by ascending timestamps, denoted as $\mathcal{G}(1 : \tau) = (\mathcal{G}_1, \dots, \mathcal{G}_\tau)$, where each subgraph $\mathcal{G}_\tau = (E_\tau, R_\tau, F_\tau)$ at timestamp τ consists of an entity set E_τ , a relation set R_τ , and a fact set F_τ . Each fact $f \in F_\tau$ is represented as a quadruple (s, r, o, τ) , where $s, o \in E_\tau$ and $r \in R_\tau$, indicating that relation r holds between entities s and o at time τ .

The primary task of TKGs extrapolation is the inference of future graph components based on observed history records $\mathcal{G}(1, \tau)$. Specifically, given a subject entity s and a relation r at a future timestamp t , the target entity o is to be predicted in order to complete the query $(s, r, ?, t)$, where $t > \tau$.

3.2. Overview of HTCGAT

To address the extrapolation reasoning task on TKGs, especially for new entities where sparse information causes difficulties in capturing causal relationships, as well as the lack of explicit modeling of causal dependencies, the Latent History Learning for Temporal Causal Graph Attention Network (HTCGAT) model is introduced in this section. The overall architecture is illustrated in Fig. 2. The model comprises four primary components: (1) Event Encoder (EE) employs a Relational Graph Convolutional Network (RGCN) to encode entities s, o and their semantic dependencies r at each timestamp t within the TKGs. (2) Latent History Learning (LHL) integrates entity similarity and temporal decay factors as a scoring mechanism to capture latent history interactions of cold-start entities to alleviate challenges in capturing their causal relationships. (3)

Temporal Causal Graph Attention Network (TCGAT) integrates Granger causality tests and temporal embeddings to construct causal weights and temporal functions. These components are embedded in the graph attention mechanism (GAT) to improve identification of causal effects and event triggers, enhancing temporal reasoning capabilities. (4) Decoder employs a Multi-Layer Perceptron (MLP) to compute the probabilities of target entities, thus enabling the prediction of missing entities.

3.3. Event encoder

At each timestamp t , entities within the subgraph exhibit interdependent relationships. To address these dependencies, a Relational Graph Convolutional Network (RGCN) is employed as the event encoder to learn embeddings for each entity at timestamp t . The method relies on the relational context of entities within the graph structure, enabling effective capture of semantic relationships and node embeddings across time, as described in embedding update formula (1):

$$h_{s,t}^{l+1} = \sigma \left(\sum_{r \in R_t} \sum_{u \in \mathcal{N}_{s,t}^r} \frac{1}{c_{s,r}} W_r^l c + W_o^l h_{s,t}^l \right) \quad (1)$$

where $\mathcal{N}_{s,t}^r$ denotes the set of neighbor nodes of node s under relation type r , $c_{s,r}$ is a normalization constant specific to node s and relation r , $h_{s,t}^l$ and $h_{o,t}^l$ represent the l layer embeddings of entities s and o at timestamp t , respectively, W_r^l and W_o^l are learnable weight matrices, and σ is the activation function.

To extract the global semantic information of a time slice from node-level representations, a pooling operation is applied to aggregate all node embeddings into a unified global embedding vector. The global embedding H^t is given as follows:

$$H^t = \frac{1}{|E_t|} \sum_{v \in E_t} h_{v,t}^{l+1} \quad (2)$$

here $|E_t|$ denotes the total number of nodes in the graph at time t .

The aggregated sequence of global embeddings $\{H^1, H^2, \dots, H^t\}$ are further organized into a variable-length sequence (Packed Sequence) for generating the subject entity s and relation type r of queries at future timestamps.

3.4. Latent history learning

In TKGs, the set of entities changes over time. A new entity w often appears at timestamp t , which makes capturing their causal relationships using history information difficult. To address this issue, we propose a Latent History Learning module. It integrates entity similarity and temporal influence to select the top- k most relevant entities as the latent history of the new entity w . These latent histories provide support for capturing causal relationships between the new entity w and other entities.

3.4.1. Relevance score between entities

The new entity w appearing at timestamp t has no history interactions. To extract the latent history information of the new entity w , we model the correlation between the new entity and history entities using cosine similarity. The number of candidate entities considered is: $N = \sum_{\tau=t-L}^{t-1} n_\tau$, where n_τ denotes the number of entities in $\mathcal{G}_\tau$, and L is the length of the history sequence. The candidate entity set is defined as $E = \{e_{1,t-L}, \dots, e_{h,\tau}, \dots, e_{n_{t-1},t-1}\}$, where $e_{h,\tau}$ denotes the entity h existing at timestamp τ . Let the feature vectors of entity w and entity h at times t and τ be $\mathbf{h}_{w,t}$ and $\mathbf{h}_{h,\tau}$, respectively. The formula for calculating cosine similarity is:

$$\cos_sim(w_t, h_t) = \frac{\mathbf{h}_{w,t} \cdot \mathbf{h}_{h,\tau}}{\|\mathbf{h}_{w,t}\| \|\mathbf{h}_{h,\tau}\|} \quad (3)$$

This formula measures the similarity between the new entity w and candidate entities in the TKGs. A higher value indicates higher similarity in their features.

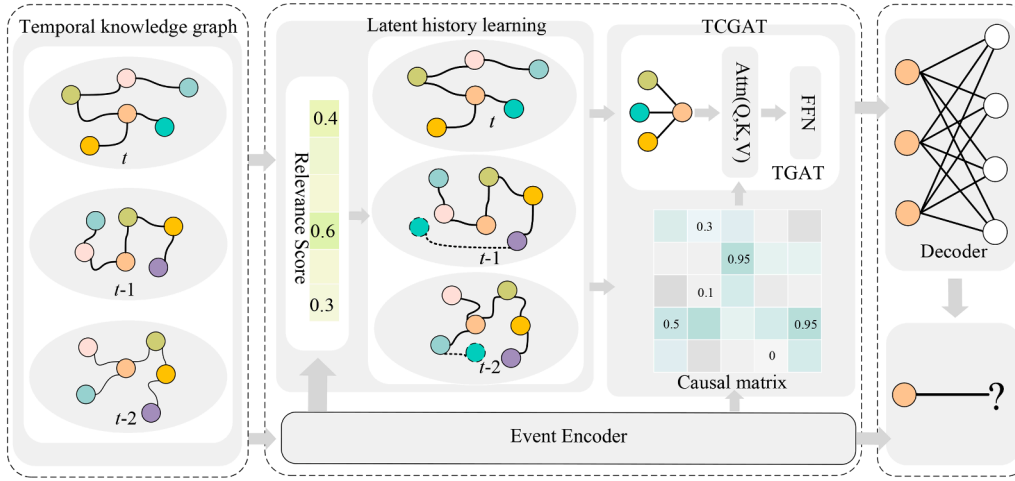


Fig. 2. The proposed HTCGAT. Latent History Learning (LHL) uses a novel scoring mechanism to learn the latent history of new entities. Temporal Causal Graph Attention Network (TCGAT) for Temporal Causality Discovery builds causal weights with the help of Granger causality test method and integrates it into Temporal Graph Attention Network (TGAT).

In TKGs, nodes that are closer in time to the current timestamp usually have a greater influence on the new entity. To more accurately mine the latent history of new entities, this paper introduces a temporal decay factor based on entity similarity to weight the importance of history entities with respect to time. Let the time weight function $w(t)$ be such that the shorter the time interval between two nodes, the greater the weight. The time weight function is specified as an exponential decay function, represented as follows:

$$w(t, \tau) = e^{-\eta|t-\tau|} \quad (4)$$

where t is the timestamp of entity v , τ is the timestamp of the candidate history node, and η is a hyperparameter controlling the speed of temporal decay.

Based on the above entity similarity and time weight, the relevance score between the history entity and the new entity can be computed as:

$$\text{score}(w_i, h_i) = \cos_sim(w_i, h_i) \cdot w(t, \tau) \quad (5)$$

A higher score indicates that the node is more relevant to the new entity in both the feature and temporal dimensions.

3.4.2. Filter the latent history

The new entity w lacks history interaction records, which hinders the extraction of its causal relationships with other entities. To address this challenge, we identify the top- k most relevant candidate entities as the latent history of w based on relevance scores. This enables the model to infer potential causal relationships indirectly, even in the absence of direct history data.

The Latent History Learning module computes relevance scores for all candidate entities. It then selects the top- k highest-scoring entities, denoted as $\{e_{1,t_1}, e_{2,t_2}, \dots, e_{k,t_k}\}$, to form the latent history set for w . To derive latent history information, past interactions involving these entities (i.e., quadruples where they appear as head or tail entities) are extracted, forming the set $H(w)$. Formally, $H(w)$ is defined as:

$$H(w) = \{(u, r, v, t_j) \mid e_{j,t_j} \in \{u, v\}, t_j < t\} \quad (6)$$

where u, v represents the entities involved in the interaction, r is the interaction relation, and t_j denotes the timestamp when the latent entity e_{j,t_j} interacted.

3.5. Temporal causal graph attention network

TKGs reasoning requires modeling complex causal dependencies and temporal event patterns. To capture these dynamics, we propose the

Temporal Causal Graph Attention Network (TCGAT), which integrates Granger causality tests and temporal embeddings into a Graph Attention Network (GAT) framework. By computing causal weights and temporal functions, TCGAT enhances the identification of causal effects and event triggers, thereby improving temporal reasoning performance.

3.5.1. Granger causality test

Significant causal dependencies exist between entities in TKGs. The state change of one entity may result from the past actions or states of others. Accurately capturing these causal relationships enables the model to make more precise predictions. The Granger causality test (Yin and Barucca (2022)) is a classic method for identifying causal relationships using temporal data (Xun et al. (2022)). Directly applying it to high-dimensional entity embeddings carries the risk of the curse of dimensionality and overfitting, while potentially masking critical dynamic patterns. To address this, we use PCA (Chang et al. (2018)) to reduce the embedding sequence to one dimension and extract the main trend. We ensure stationarity by applying the ADF (Dickey and Fuller (1976)) test and perform differencing when necessary. Finally, we construct a linear regression using the processed entity embedding sequence. The causal strength between entity pairs is measured by statistical significance (p-value). The specific process is as follows:

When calculating causal weights using the Granger causality test, we utilize history embedding information to identify causal dependencies between entities. For each interaction in the training set, the history embedding trajectory of entity pairs s, o is constructed as:

$$\chi_{s,t} = \{h_{s,t-L}, \dots, h_{s,t-1}\}, \quad \chi_{o,t} = \{h_{o,t-L}, \dots, h_{o,t-1}\} \quad (7)$$

where L is the lag window length, $h_{s,t-L}$ and $h_{s,t-1}$ represent the embedding of node s at times $t-L$ and $t-1$, $h_{o,t-L}$ and $h_{o,t-1}$ represent the embedding of node o at times $t-L$ and $t-1$.

Following PCA-based dimensionality reduction, we test each dimension of the time series for stationarity using the ADF test. Non-stationary series are differenced independently for each dimension. The resulting series are then re-examined, with the procedure repeated up to two times. If a series remains non-stationary after two rounds of differencing, the final differenced series is used for Granger causality analysis. This results in stationary one-dimensional scalar time series, denoted as:

$$\hat{\chi}_{s,t} = \{\hat{h}_{s,t-L}, \dots, \hat{h}_{s,t-1}\}, \quad \hat{\chi}_{o,t} = \{\hat{h}_{o,t-L}, \dots, \hat{h}_{o,t-1}\}. \quad (8)$$

The embeddings $\hat{\chi}_{s,t}$ and $\hat{\chi}_{o,t}$ are concatenated into a two-dimensional time series data. From this data, the following two linear

regression models are constructed. The lagged model incorporates the history information of the other entity to test whether it improves prediction performance, while the non-lagged model serves as a baseline that uses only the entity's own history. This comparison helps determine whether a significant Granger causal relationship exists between the entities.

Non-lagged model (no causal relationship):

$$\hat{\chi}_{o,t} = \alpha + \sum_{k=1}^L \beta_k \hat{\chi}_{o,t-k} + \epsilon_t \quad (9)$$

where α is the constant term, β_k is the regression coefficient, L is the lag order of the regression, and ϵ_t is the error term.

Lagged model (causal relationship exists):

$$\hat{\chi}_{o,t} = \alpha + \sum_{k=1}^L \beta_k \hat{\chi}_{o,t-k} + \sum_{k=1}^L \gamma_k \hat{\chi}_{s,t-k} + \epsilon_t \quad (10)$$

where γ_k is the regression coefficient.

If the lagged model significantly outperforms the non-lagged model (with a smaller p-value obtained from an F-test), Granger causality is considered to exist between entity pairs (s, o) at time t . Due to the low significance level, we use a mapping function to constrain the causal weights within a reasonable range. The causal weight between any two entities s and o at time t is defined as:

$$w_{(s,o)}^t = \begin{cases} \frac{1}{1+p\text{-value}}, & \text{if } p\text{-value} < \delta \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

where $w_{(s,o)}^t$ represents the strength of the prediction capability of entity s for entity o at timestamp t , p-value is the significance level result of the Granger causality test, and δ is the threshold value.

The collection of these causal weights forms the causal matrix $W = \{w_{(s,o)}^1, w_{(s,o)}^2, \dots, w_{(s,o)}^t\}$.

3.5.2. Time embedding

In TKGs, the evolution of events depends not only on the structural relationships and causal dependencies between entities but also heavily on the specific timestamps of events and the influence of their relative time intervals. Properly incorporating temporal information is crucial for capturing the dynamic evolution of entities and enhancing the model's temporal modeling capabilities. To this end, the TCGAT model in this paper adopts a kernel-based time embedding method. This method encodes the relative time differences into a low-rank vector representation using a kernel function, allowing the model to better capture the impact of time order and time intervals on the occurrence of events.

We use a cosine kernel function to encode the relative time difference between entity s and entity o :

$$\varphi(t - \tau) = \cos(\omega(t - \tau) + b) \quad (12)$$

where w and b are trainable parameters, t and τ are the timestamps corresponding to entities s and o , respectively.

3.5.3. TCGAT

In TKGs, the evolution of entities depends not only on their own state but also on causal relationships and temporal dependencies. Traditional Graph Neural Networks (GNNs) are typically built on static adjacency structures, making it difficult to reveal the dynamic evolution of causal structures and to capture the time-varying influence between entities. To address this, this paper proposes the Temporal Causal Graph Attention Mechanism (TCGAT), where the attention mechanism incorporates a causal relationship matrix and time weights when computing attention between nodes. This fusion approach enables the model to identify and strengthen key causal relationships and ensure its sensitivity to temporal dependencies. The specific process of the Temporal Causal Graph Attention Mechanism is as follows:

Given a source node s and neighboring nodes $\{1, 2, \dots, n\}$, along with timestamps $\{\tau_1, \tau_2, \dots, \tau_n\}$, the model generates the query vector $\mathbf{Q}$, key vectors $\mathbf{K}$, and value vectors $\mathbf{V}$ based on node features and time embeddings.

The query vector $\mathbf{Q}$ is computed from the source node's features and temporal embedding, while key vectors $\mathbf{K}$ and value vectors $\mathbf{V}$ are derived from neighbor features and their respective time embeddings:

$$\mathbf{Q} = (h_{s,\tau} \parallel r_0 \parallel \varphi(0)) \cdot W_q \odot W \quad (13)$$

$$\mathbf{K} = \left[(h_{1,\tau_1} \parallel r_{s,1} \parallel \varphi(t - \tau_1)), \dots, (h_{n,\tau_n} \parallel r_{s,n} \parallel \varphi(t - \tau_n)) \right] \cdot W_k \quad (14)$$

$$\mathbf{V} = \left[(h_{1,\tau_1} \parallel r_{s,1} \parallel \varphi(t - \tau_1)), \dots, (h_{n,\tau_n} \parallel r_{s,n} \parallel \varphi(t - \tau_n)) \right] \cdot W_v \quad (15)$$

where $W_q, W_k, W_v \in \mathbb{R}^{d_{h+r+\varphi} \times d}$ are the corresponding weight matrices, $W \in \mathbb{R}^d$ is the causal matrix, r_0 is the initial feature vector of the node s , and $\varphi(0)$ is the initial time embedding feature. h_{n,τ_n} represents the feature vector of node n at time τ_n , $r_{s,n}$ represents the edge feature between nodes s and n , and $\varphi(t - \tau_n)$ represents the time embedding feature between nodes s and n .

The attention weight is computed based on the dot product of the query ($\mathbf{Q} \in \mathbb{R}^{1 \times d}$) and the key ($\mathbf{K} \in \mathbb{R}^{n \times d}$):

$$\alpha = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right) \quad (16)$$

where d is the dimension of the key vector, and the function softmax is used to convert the dot product result into a probability distribution.

The output vector is computed through an attention-based weighted summation of the value vectors ($\mathbf{V} \in \mathbb{R}^{n \times d}$):

$$h_s(t) = \alpha \cdot \mathbf{V} \quad (17)$$

In TKGs, the output vector represents the influence of neighboring nodes $\{1, 2, \dots, n\}$ on the source nodes s , and also represents the relationship between source nodes s and neighboring node $\{1, 2, \dots, n\}$.

Entity dependencies in multi-entity temporal modeling exhibit significant dynamics and heterogeneity. Single attention heads fail to capture evolving interaction patterns. This leads to information bottlenecks or underfitting and restricts model performance. To mitigate this, the basic attention mechanism with causal weights and temporal embeddings is extended to a multi-head structure. Temporal embeddings and causal weights are processed independently by each attention head; their outputs are concatenated with the source node's original features and passed to a feedforward neural network. This design improves node representation quality and enhances model generalization.

Specifically, the output vector is concatenated with the source node's features and passed through a feedforward neural network (FFN) to obtain the output of the l-layer skip connection:

$$h_s^l(t) = \text{FFN}(h_s^1(t) \parallel h_s^2(t) \parallel \dots \parallel h_s^k(t) \parallel h_{s,\tau}) \quad (18)$$

where $h_s^k(t)$ is the output vector of the k-th head, and $h_{s,\tau}$ is the original feature vector of source node s .

3.6. Parameter learning

This section describes the training process of HTCGAT to perform event inference across multiple timestamps. Parameter learning is carried out through an event prediction task. Given a subject entity s and a relation r , the task of predicting the target entity is formulated as a multi-class classification problem, where each class corresponds to a possible target entity. In the HTCGAT framework, the probability formula for the target entity o can be represented as follows:

$$p(o_t | s_t, r_t) = \text{softmax}\left([h_{s,t} : h_s^l(t)]^T \cdot W_{O_t}\right) \quad (19)$$

where $h_{s,t}$ is the static embedding of the source node s . $h_s^l(t)$ is the dynamic local representation generated by TCGAT, capturing the temporal

features of the original entity s and relation r . W_{O_i} represents the parameters of the MLP decoder.

we define the corresponding loss function. For the event prediction task at each timestamp, the loss function can be represented as:

$$\mathcal{L}_{\text{task}} = - \sum_{(s,r,o,t) \in F_t} \log(p(o_t | s_t, r_t)) \quad (20)$$

The HTCGAT model introduces the Granger causality test to construct causal weights, and the loss function for causal weights can include a regularization term to ensure the validity of the causal weights. For example, sparsity regularization can be used to constrain the causal weights, avoiding overfitting. Specifically:

$$\mathcal{L}_{\text{causal}} = \sum_i \sum_j \|w_{i,j}\|_1 \quad (21)$$

The final loss function:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \mathcal{L}_{\text{causal}} \quad (22)$$

where λ is the parameter controlling the importance of the causal loss term

4. Experiments

4.1. Datasets and evaluation metrics

To validate the effectiveness of the proposed method, three commonly used datasets for TKGs reasoning tasks are employed: ICEWS18 (Boschee et al. (2018)) and ICEWS05-15 (Boschee et al. (2015)) datasets are derived from the Integrated Crisis Early Warning System, which includes global event data from 1995 to the present. GDEL dataset (Lee-taru and Schrodt (2015)) is derived from the Global Database of Events, Language, and Tone, which covers global news reports since 1979. The ICEWS18 and ICEWS05-15 datasets record events from 2018 and 2005 to 2015, respectively, while the GDEL dataset records events from 1979 to the present. The division of the datasets is shown in Table 1.

We evaluate the performance of the HTCGAT model using standard metrics: Mean Reciprocal Rank (MRR) and Hits@{1,3}. The formal definitions of these metrics follow:

$$MRR = \frac{1}{|S|} \sum_{i=1}^{|S|} \frac{1}{rank_i} \quad (23)$$

$$Hits@K = \frac{1}{|S|} \sum_{i=1}^{|S|} \mathbb{I}(rank_i \leq K) \quad (24)$$

where S represents the set of test queries, $rank_i$ denotes the rank position of the correct answer for the i -th query, and $\mathbb{I}$ is the indicator function.

MRR measures the average reciprocal rank of correct answers. Hits@K calculates the proportion of queries where the correct answer appears in the top-K predictions: Hits@1 and Hits@3 evaluate the model's precision at the first and top three ranking positions respectively. These metrics collectively provide a rigorous assessment of the model's reasoning accuracy and ranking capability.

Table 1

Statistical overview across datasets and metrics. Each row is a dataset, and each column a metric. $|E|$, $|R|$, $|T|$ indicates the number of entities, relations and timestamps and $|F_{\text{train}}|$, $|F_{\text{valid}}|$, $|F_{\text{test}}|$ denote the number of facts in the training, validation, and test sets, respectively.

Dataset	$ E $	$ R $	$ T $	$ F_{\text{train}} $	$ F_{\text{valid}} $	$ F_{\text{test}} $
ICEWS18	23,033	256	304	373,018	45,995	49,545
ICEWS0515	10,488	251	4017	322,958	69,224	69,147
GDEL	7691	240	2976	1,734,399	238,765	305,241

4.2. Baselines

HTCGAT is evaluated against established TKGs reasoning approaches, which fall into two main categories: RNN-based architectures and graph attention-based methods. The first group of models, including RENET (Jin et al. (2020)), XERTE (Han, Chen, Ma, (2021)), RE-GCN (Li et al. (2021)), TLogic (Liu et al. (2022)), CEN (Li et al. (2022b)), Daemon (Dong et al. (2023)), CENET (Xu et al. (2023)), and RETIA (Liu et al. (2023)), use recurrent neural network architectures to model temporal sequences and capture fact evolution across consecutive timestamps. These approaches have shown competitive results across various temporal reasoning tasks.

The second category utilizes graph attention mechanisms for temporal information processing. Models such as TECHS (Lin et al. (2023)), HTCCN (Chen et al. (2024)), TiPNN (Dong et al. (2024)), CaORG (Fan et al. (2025)) and CasTAN (Zheng et al. (2025)) aggregate features using temporal context weights, improving the accuracy of time-dependent prediction modeling. Among them, HTCCN (Chen et al. (2024)) is a significant advancement as the first framework to integrate Hawkes processes with Causal Convolutional Neural Networks (CNNs). CasTAN (Zheng et al. (2025)) incorporates a CaST module that integrates spatial, temporal, and causal attention mechanisms together with an adaptive feature fusion gate. This allows the model to capture richer entity and relation representations, further improving its expressive power.

In addition, we also consider TV-LLM (Pan et al. (2025)). This work attempts to use large language models to generate temporal rules, which helps improve the accuracy of temporal knowledge graph reasoning.

4.3. Experimental settings

To ensure a fair comparison, we applied a consistent set of hyperparameters for all HTCGAT experiments. Specifically, the hidden layer size was set to 200 neurons, the Adam optimizer was used with a learning rate of 0.001, training was conducted for up to 20 epochs, dropout was set to 0.2, and the batch size was 128. The parameters λ were set to 0.01. Weight initialization and regularization strategies were kept consistent across all experiments. All experiments were conducted on an NVIDIA RTX 4090 GPU.

For RE-GCN (Li et al. (2021)), CEN (Li et al. (2022b)), RETIA (Liu et al. (2023)), HTCCN (Chen et al. (2024)), CasTAN (Zheng et al. (2025)), the embedding dimension for entities and relations is set to 200, dropout is 0.2, and the Adam optimizer has a learning rate of 0.001. All other parameters follow the settings provided in the original papers. In the Daemon (Dong et al. (2023)) and CaORG (Fan et al. (2025)) experiments, the Adam optimizer uses a learning rate of 0.0005, and Daemon's entity and relation embedding dimensions are set to 64. All other parameters follow the settings provided in the original papers. The experimental results for CENET* (Xu et al. (2023)), XERTE* (Han, Chen, Ma, (2021)), TLogic* (Liu et al. (2022)), TiPNN* (Dong et al. (2024)), TECHS* (Lin et al. (2023)), RENET* (Jin et al. (2020)), and TV-LLM* (Pan et al. (2025)) are sourced from previous papers (Han, Chen, Ma, (2021); Chen et al. (2024); Pan et al. (2025)).

4.4. Performance comparison

To verify that the HTCGAT model can capture complex causal relationships, especially under the cold-start problem and complex temporal dependencies, we compare it with baseline models by performing extrapolation on the datasets. The specific results appear in Table 2. Compared to all 14 state-of-the-art benchmarks from 2021 to 2025, HTCGAT consistently demonstrates superior performance. In the ICEWS18 and GDEL datasets, the MRR and Hits@1,3 metrics show significant improvement, but in the ICEWS05-15 dataset, the HTCGAT model improved MRR by 1.31, decreased Hits@1 by 0.86, and

Table 2

HTCGAT performs a performance comparison on three datasets in terms of MRR, Hit@1, and Hit@3.

Method	ICEWS18			ICEWS0515			GDELT		
	MRR	H@1	H@3	MRR	H@1	H@3	MRR	H@1	H@3
RE-GCN	30.55	20.00	34.73	46.41	35.17	52.76	19.31	11.99	20.61
RENET*	28.81	19.05	32.44	42.97	31.26	46.85	19.55	12.38	20.80
xERTE*	29.31	21.03	33.51	46.62	37.84	52.31	19.45	11.92	20.84
CEN	31.50	21.70	35.44	46.84	36.38	52.45	21.16	13.34	22.71
TLogic*	29.82	20.54	33.95	46.97	36.21	53.13	22.73	17.65	24.66
DaeMom	31.85	22.67	35.92	47.85	37.90	52.61	20.73	13.65	22.53
CENET*	27.85	18.15	31.63	41.95	32.17	46.93	20.23	12.69	21.70
RETIA	34.16	22.97	39.27	47.27	36.65	52.91	19.73	12.54	21.01
TECHS*	30.85	21.81	35.39	48.38	38.34	54.69	-	-	-
HTCCN	35.63	24.90	39.26	51.94	40.32	57.79	23.46	15.18	25.21
TIPNN*	32.17	22.74	36.24	-	-	-	21.17	14.03	22.98
CaORG	32.74	21.27	37.37	49.38	38.97	55.02	23.55	14.68	25.55
TV-LLM*	33.2	22.9	37.9	50.8	41.3	56.6	-	-	-
CaSTAN	33.17	26.04	35.74	47.01	48.62	49.49	11.91	6.67	12.06
HTCGAT	41.09	34.00	43.81	53.25	47.76	55.53	40.57	33.56	43.13
	5.46↑	7.96↑	4.54↑	1.31↑	0.86↓	2.26↓	17.02↑	15.91↑	17.58↑

decreased Hits@3 by 2.26. This is because the HTCGAT model captures time and causal relationships more precisely. It prioritizes predictions for the most likely answer at the current timestamp to improve Hits@1 and may reduce the likelihood for other timestamps or causal chains, thus decreasing Hits@3. A detailed analysis of the experimental results reveals that HTCGAT outperforms in the following aspects:

Temporal Dependency: Most baselines, such as RENET, XERTE, RE-GCN, CEN, Daemon, CENET, and RETIA, use RNNs to consider facts at adjacent timestamps and perform well in experiments. These models capture some complex temporal dependencies but face problems such as vanishing or exploding gradients and limited computational efficiency. They also struggle to capture long-range dependencies effectively. HTCGAT mitigates the limitations of traditional RNN models by introducing a temporal embedding function in the graph attention mechanism. We also consider TV-LLM (Pan et al. (2025)), leveraging large language models to generate temporal rules and enhance temporal knowledge graph reasoning. HTCGAT, in comparison, shows superior performance in causal modeling, long-term temporal dependency capture, and robustness. TV-LLM offers more flexibility, but its stability and interpretability are lower.

Interpretability: Graph attention mechanisms are used by TECHS, CaORG, CaSTAN and TIPNN to process temporal information. The core of TIPNN lies in learning query-related path features via multi-layer aggregation, including temporal path layers, providing structural interpretability. The core of CaORG lies in constructing candidate relation graphs to effectively capture complex relational patterns for query-candidate node pairs. It demonstrates strong capabilities in local evidence integration and logical reasoning. When graph structures are sparse or incomplete, the performance and interpretability of these methods degrade significantly. These models often neglect the potential causal dependencies between entities. HTCCN, as the first model combining Hawkes processes with convolutional neural networks, advances causal and extrapolative reasoning. Limitations remain in HTCCN's ability to model global temporal dependencies, particularly long-term causal relationships across multiple time steps. CaSTAN incorporates the CaST module. It combines spatial, temporal, and causal attention mechanisms with an adaptive feature fusion gate. The model captures local spatio-temporal-causal relationships effectively. It struggles with sparse graphs and long-term causal dependencies. Computational cost is high. Interpretability is limited. The proposed HTCGAT uses a Temporal Causal Graph Attention Network to capture latent temporal information and causal dependencies between entities.

It outperforms most benchmark methods and effectively models fact evolution.

4.5. Ablation study

Using the MRR evaluation metric, ablation experiments were conducted on all three TKG datasets to investigate the effectiveness of the model. The main focus is to analyze the effectiveness of two key design modules: Latent History Learning (LHL) and Temporal Causal Graph Attention Network (TCGAT). The experimental results are shown in Fig. 3, which presents a comparative evaluation of four sub-models: (1) Our proposed method, denoted as “HTCGAT” (2) HTCGAT with TCGAT replaced by LSTM, denoted as “LHL-lstm”, (3) HTCGAT with TCGAT replaced by Temporal Graph Attention Network (TGAT), denoted as “LHL-TGAT”, and (4) HTCGAT with the LHL removed, denoted as “TCGAT”.

The experimental results presented in Fig. 3 reveal that LHL-lstm, LHL-TGAT, and TCGAT all demonstrate substantial reductions in MRR performance compared to HTCGAT. These findings validate that the integrated LHL and TCGAT model causal dependencies and evolutionary patterns successfully. This leads to notable improvements in reasoning accuracy for TKGs.

Ablation studies show consistent MRR score degradation across all sub-model variants compared to the full HTCGAT framework. The performance gap between “LHL-lstm” and “LHL-TGAT” highlights the importance of causal relationship modeling for reasoning accuracy. Comparison between “HTCGAT” and “TCGAT” demonstrates that combining cosine similarity with temporal decay effectively addresses the cold-start entity problem by enabling causal relationship learning for new entities through latent representations.

4.6. Hyperparameters sensitivity

To better understand the impact of key hyperparameters on model performance, we conducted a sensitivity analysis, focusing on three aspects: (1) the parameter η in the time decay function (Eq. (4)); (2) the selection of historical top-k; and (3) the regularization weight λ (Eq. (22)).

Hyperparameter Tuning: We used a grid search strategy for systematic tuning. The search ranges were as follows: $\eta \in \{0.01, 0.005, 0.001\}$, $k \in \{0, 2, 4, 6, 8, 10\}$, and $\lambda \in \{0, 0.00001, 0.0001, 0.001, 0.005, 0.01\}$. For each configuration, we ran the model five times using three different random seeds and took the average performance. The parameters that performed best were then used for the final test.

Effect of η : The hyperparameter η determines how quickly historical information diminishes as the time gap increases. In the ICEWS18 dataset, the timestamps are spaced 24 hours apart. We tuned the hyperparameters based on this fixed interval. Different k-values were also considered. In this way, we examined how the number of latent histories learned under different η settings affects model performance. Fig. 4 shows that larger η values make historical information decay quickly. More histories with larger k-values do not help. The model still fails to capture long-term dependencies. Performance drops significantly. Smaller η values make the model include too much long-term history. With larger k, irrelevant information enters the model. This slightly reduces performance compared to the best setting. The overall effect remains small. Overall, across all ICEWS datasets, a value of 0.005 for η combined with a moderate k gives the most stable performance. For the GDELT dataset, we used 0.007 for η based on similar considerations.

Effect of k: The k-value determines the number of latent histories learned by cold-start entities. We use the MRR evaluation metric to study the impact of different k-values on reasoning accuracy across all three TKG datasets. As shown in Fig. 5, the performance of the model under different k-values is explained: In the ICEWS18 dataset, reasoning performance reaches its peak when k is between 4 and 8; in the ICEWS05-15 dataset, reasoning performance peaks when k equals 6; in the GDELT dataset, reasoning performance reaches its peak when k equals 4. The performance of the HTCGAT model initially improves with the increase

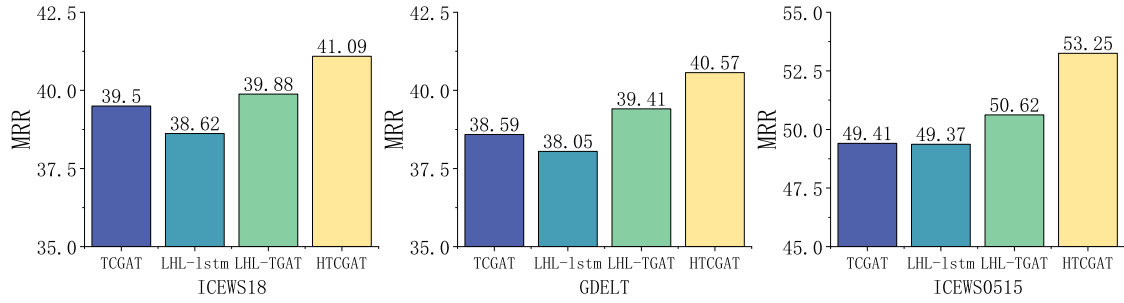


Fig. 3. Ablation experiments of the proposed HTCGAT model. By systematically removing the two main modules, TCGAT and Latent History Learning, the reasoning performance of HTCGAT showed varying degrees of decline.

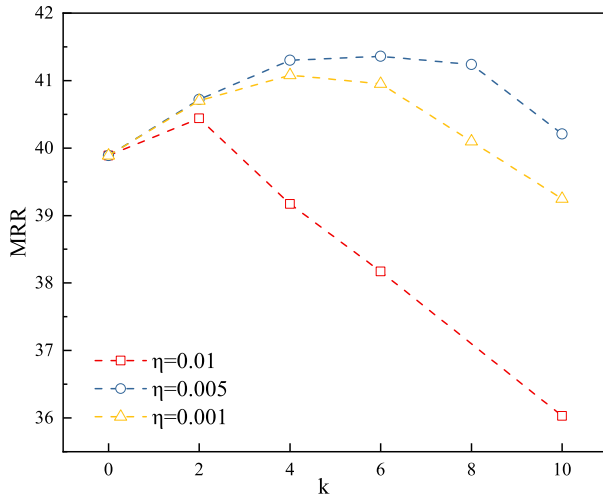


Fig. 4. MRR performance on ICEWS18 with varying history decay rate η and top- k selection size.

of k -value. This suggests that the latent history learned by HTCGAT provides more effective information for TKG reasoning. As the k -value increases, the model's performance declines. This indicates that the latent history learned by HTCGAT contains much irrelevant information, which interferes with the model's reasoning.

Effect of λ : To understand how λ influences model performance, we performed an ablation study on the ICEWS18 dataset, as illustrated in Fig. 6. The experimental results suggest that the model exhibits limited sensitivity to the regularization parameter λ , likely because the causal weights are initialized based on Granger causality. When λ is increased from 0 to 0.001, an improvement of approximately 0.3 percentage points in MRR can be observed, indicating that moderate regularization may help improve prediction accuracy. If λ becomes too large, the model tends to rely excessively on the regularization term, which can lead to performance degradation. Overall, these observations highlight the robustness of HTCGAT to variations in λ and offer practical guidance for its selection.

4.7. Experimental stability under PCA dimensionality reduction

As shown in Table 3, to clearly illustrate the impact of PCA-based dimensionality reduction on the time cost and model performance of the HTCGAT model, we provide a comparison of embeddings reduced to different dimensions. When comparing different PCA dimensions, using higher dimensions makes HTCGAT take much longer to preprocess and train. The MRR and Hit@1 scores stay almost the same, changing by less than 0.5 %. This suggests that higher-dimensional embeddings offer

Table 3

Comparison of Different PCA Reduced Dimensions on HTCGAT Model (ICEWS18 Dataset).

Embedding Dimension	Pre-training Time (s)	Training Time (s)	MRR	Hit@1
PCA 5D	40	272	41.23	34.38
PCA 3D	34	241	41.33	34.51
PCA 1D	24	220	41.36	34.62

Table 4

Comparison of different pooling strategies on ICEWS18.

Pooling Method	MRR	Hit@1	Hit@3
Attention	41.36	34.62	43.86
Max	41.21	34.49	43.88
Mean	41.27	34.40	43.97

little gain but add computational cost. Lower-dimensional PCA, like 1D, is more efficient and practical.

4.8. The impact of different pooling methods on experiments

In HTCGAT, the global embedding H_t is obtained by aggregating node embeddings using mean pooling. Its main role is to produce entity and relation representations for later queries. Its effect on overall model performance is relatively small. To test the validity of this aggregation strategy, three alternative pooling methods were used: mean pooling, max pooling, and attention-based pooling. The experimental results on the ICEWS18 dataset are shown in Table 4.

The experimental results show that different pooling strategies have a negligible effect on overall model performance. Attention-based pooling provides only slight improvements over mean and max pooling. This suggests that simple average pooling is sufficient to capture the essential information of global temporal representations. It also preserves reliable query generation. Overall, the choice of global embedding pooling strategy has little impact on performance. This provides empirical support for the approach used in our work.

4.9. Comparison of different p -value mappings

In our proposed HTCGAT model, the computation for the Granger causality test is defined by Eq. (11). This formula is not the only theoretical choice. To verify its validity, we also implemented an alternative approach that maps causal weights using the negative logarithm of the p -value.

As shown in Table 5, the results indicate that the two mapping methods have similar effects on model performance: MRR values are 41.18 and 41.36, Hit@1 values are 34.33 and 34.62, and Hit@3 values are 43.95 and 43.86, respectively. The result shows that the choice of

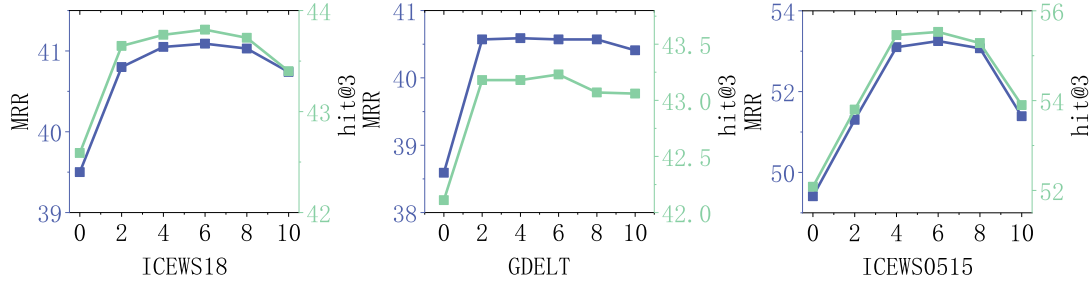


Fig. 5. The performance of HTCGAT in terms of MRR and Hits@3 for different values of k .

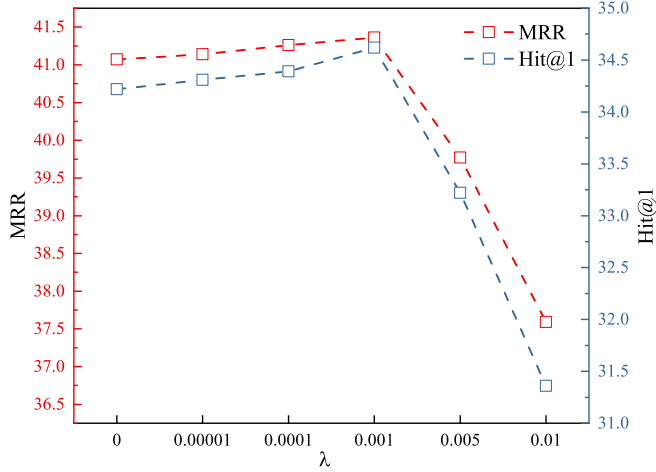


Fig. 6. Performance comparison with varying regularization strength λ on ICEWS18.

Table 5
Comparison of different Granger mapping methods on ICEWS18.

Method	MRR	Hit@1	Hit@3
$-\log(p)$	41.18	34.33	43.95
Eq. (11)	41.36	34.62	43.86

Eq. (11) is reasonable. It performs reliably in practice and effectively guides the application of causal weights.

4.10. Performance evaluation of the model under data scarcity

In real-world scenarios, training data is often limited or unevenly distributed, which can impact the model's ability to generalize. Therefore, it is important to evaluate the model's robustness under conditions of limited data. In this section, we examine the HTCGAT model under conditions of limited training data. To do so, we systematically reduce the training set by 10%, 20%, and 50%. This allows us to assess the model's stability and adaptability under both moderate and extreme data scarcity.

As shown in the Table 6, reducing the training set by 10% or 20% leads to similar drops in HTCGAT's performance on the test set. This suggests that the model is robust when a small portion of the data is removed. When the training set is reduced by 50%, HTCGAT's performance drops noticeably. This suggests that the model struggles to achieve strong generalization when half of the data is available. Overall, the model maintains high accuracy under mild data scarcity, indicating that it is both robust and capable of generalizing well.

Table 6

Comparison of ICEWS18 Dataset Performance after Removing Different Proportions of Data.

Training Data Reduction Ratio	MRR	Hit@1	Hit@3
0	41.36	34.62	43.86
10%	40.83	33.97	43.55
20%	40.33	33.48	42.92
50%	37.67	30.83	40.32

4.11. Case study

Fig. 7 shows three query results inferred by the proposed HTCGAT model, with North Korea as the main focus.

(North Korea, negotiate, $?, t$): The correct answer is South Korea. At times $t-3$ and $t-2$, there were multiple visits and interactions between North Korea and South Korea, indicating a close diplomatic relationship and interaction (cause). The HTCGAT model assigned high causal attention weights to these visit events, indicating that it identified them as the main contributors to the subsequent negotiation at time t (effect). The HTCGAT model successfully inferred this relationship, validating its effectiveness in capturing causal relations among entities with frequent interactions.

(North Korea, visit, $?, t$): The correct answer is Russia. In the history interaction data, Russia appeared at time $t-1$, with no prior history interactions. The Russian diplomat is semantically highly related to Russia. HTCGAT incorporated the semantic similarity between Russian diplomats and Russia into the reasoning process and assigned higher attention to these relevant features. As a result, it correctly inferred that North Korea would choose to visit Russia. This indicates that the model not only captures explicit relations but also effectively identifies latent causal dependencies between new and existing entities, such as between Russia and North Korea.

(North Korea, consult, $?, t$): The correct answer is Trump. Trump signed an agreement with North Korea at time $t-3$, despite controversies at times $t-2$ and $t-1$, and again signed a new agreement at time $t-1$. This laid the groundwork for both sides' willingness to engage in dialogue (cause). HTCGAT assigned high causal attention to these past interactions. This allowed the model to correctly infer that North Korea would consult Trump at time t (effect). This query demonstrates that the HTCGAT model can accurately capture complex causal chains over long temporal spans, enabling cross-timepoint reasoning.

These queries illustrate the proposed HTCGAT model's ability to identify complex causal relationships and make more reasonable and accurate predictions.

4.12. Computational complexity of HTCGAT

The proposed HTCGAT model does not recalculate Granger causality weights at each training iteration, as this would lead to excessive computational overhead. Due to the dynamic evolution of causal relationships among entities in temporal knowledge graphs, the causality weights

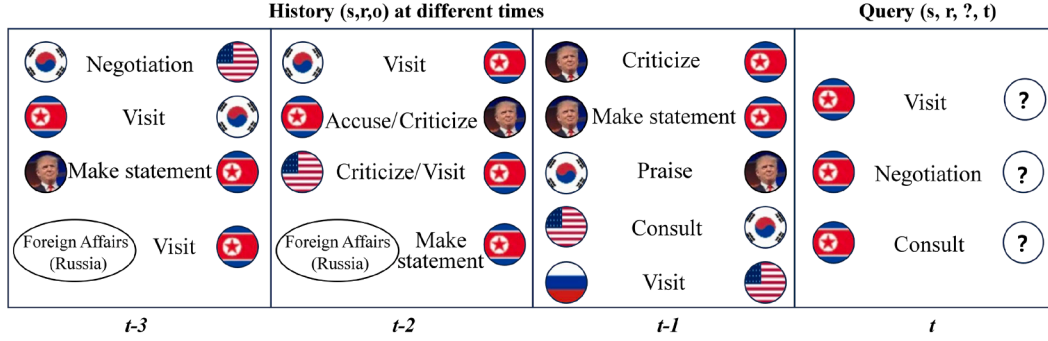


Fig. 7. HTCGAT Prediction Case Study. we select three queries centered around North Korea for analysis.

are not completely fixed. In the pre-training phase, HTCGAT computes initial causal weights from historical data using Granger causality tests to serve as prior knowledge for subsequent training. When new data arrive, such as an increase in the number of entities or the addition of new time slices, HTCGAT recalculates the causal weights using Granger causality tests to reflect the updated causal relationships among entities. During normal training, these initial weights are regarded as trainable parameters and are adjusted gradually so that the model can adapt to the evolving causal dependencies over time.

From the perspective of computational complexity, the core operation in the pre-training phase is performing Granger causality tests on each training quadruple. We denote by N the number of training tuples, by L the maximum lag period, by E the number of entities, and by D the dimensionality of the entity embeddings. For each quadruple, we first construct historical embedding sequences of length L for the two entities, and then perform a Granger causality test between them, requiring $O(L)$ and $O(L^3)$ time, respectively. This results in a total time complexity of $O(N \cdot L^3)$. The space overhead primarily arises from caching the embeddings of E entities over the last L time steps, resulting in a space complexity of $O(N + E \cdot L \cdot D)$.

HTCGAT utilizes the temporal knowledge graph together with a pre-trained causal weight matrix during training and inference, computing node representations through a temporal graph attention network. At each layer, the model recursively samples K neighbors for each node to compute feature representations. For layer l , the number of nodes is approximately $B \cdot K^{l-1}$, where B represents the batch size. Each node performs a linear transformation followed by a neighbor aggregation operation, which together require $O(H^2)$ time, where H is the hidden dimension. Consequently, the time complexity of layer l is $O(B \cdot K^l \cdot H^2)$. Extending this to a network with n_{layers} layers, the total time complexity becomes $O(B \cdot K^{n_{\text{layers}}} \cdot H^2)$, and the corresponding space complexity is $O(B \cdot K^{n_{\text{layers}}} \cdot H)$. The element-wise application of causal weights in attention computation only adds a constant-order overhead of $O(B \cdot K)$ time and $O(N)$ space, which can be ignored.

In summary, the computational bottleneck of HTCGAT is confined to the pre-processing phase, whereas both its training and inference stages exhibit computational complexity on par with standard GNN baselines. This approach leverages an offline, one-time investment in causal analysis to enable efficient online operation, striking a balance between model capacity and practical scalability.

5. Conclusion and future work

The proposed Latent History Learning for Temporal Causal Graph Attention Network (HTCGAT) addresses two core challenges in TKG reasoning: capturing complex temporal-causal dependencies and the difficulty of modeling causal relationships for cold-start entities. It infers latent histories for cold-start entities by using entity similarity and temporal decay to enable more accurate causal modeling. HTCGAT integrates Granger causality testing with Temporal Graph Attention Net-

works (TGAT) to identify causal dependencies and event-triggering patterns, improving prediction performance. Experimental results demonstrate superior accuracy in temporal entity prediction, confirming the model's effectiveness in causal modeling. Two limitations remain: (1) high computational complexity reduces training efficiency, and (2) reliance on predefined graph structures limits causal detection accuracy. Future work will explore more efficient graph convolution architectures to handle large-scale temporal data and incorporate domain knowledge to optimize graph construction for better inference.

CRedit authorship contribution statement

Jianghui Cai: Writing – review & editing, Methodology, Investigation, Conceptualization; **Cuicui Xu:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization; **Haifeng Yang:** Writing – review & editing, Methodology, Conceptualization; **Xin Chen:** Writing – review & editing, Methodology, Conceptualization; **Aiyu Zheng:** Data curation, Formal analysis, Methodology; **Jing Hao:** Conceptualization, Supervision, Project administration; **Yaling Xun:** Validation, Investigation, Formal analysis; **Xujun Zhao:** Resources, Software.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Attention Visualization

To demonstrate the role of causal attention in HTCGAT, we provide visualizations of two queries drawn from the test set of the ICEWS18 dataset. These examples make clear which historical events exert the greatest influence on the prediction of subsequent interactions.

As show in Fig. A.1, we selected a representative query (Foreign Affairs (Holy See (Vatican City State)), Criticize or denounce, ?, t) and visualized a typical reasoning path. Consider the path Foreign Affairs (Holy

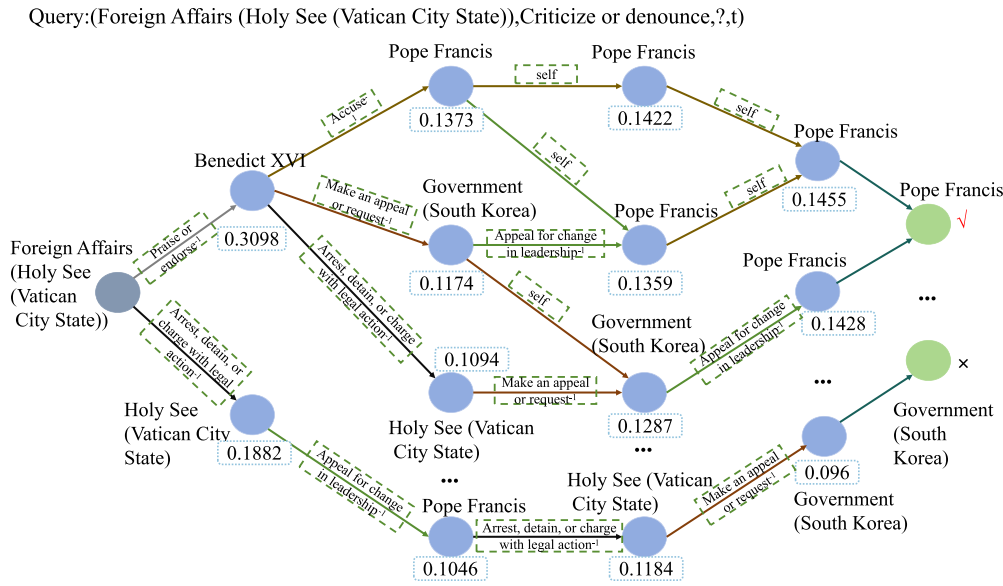


Fig. A.1. Visualization Case Study of HTCGAT.

See (Vatican City State)) → Benedict XVI → El-Ghad → Government (South Korea) → Pope Francis as an example. The attention weights along its edges are 0.3098, 0.1094, 0.1287, and 0.1428. The cumulative attention score reaches approximately 6.23×10^{-4} . Visualization shows that attention propagates stepwise along the path. As shown in the path, Pope Francis criticized Benedict XVI, and Benedict XVI then praised and endorsed the Foreign Affairs (Holy See (Vatican City State)) (cause). This behavior leads to Foreign Affairs (Holy See (Vatican City State)) criticizing or condemning Pope Francis (effect). This demonstrates that the model is effective in capturing the inherent causal relationships.

These examples show that HTCGAT's causal attention weights reflect meaningful, domain-relevant causal relations, making the learned temporal dynamics easier to understand.

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